Configuração 6: $M$ = 128, $C$ = 8, $K$ = 8, IoT = 20 dB, QPSK

Perda de SNR do BCD-MMSE (iter 4) em SER = $10^{-2}$, em dB, em relação à referência ideal (float64); × = não chega a $10^{-2}$ nas SNRs simuladas

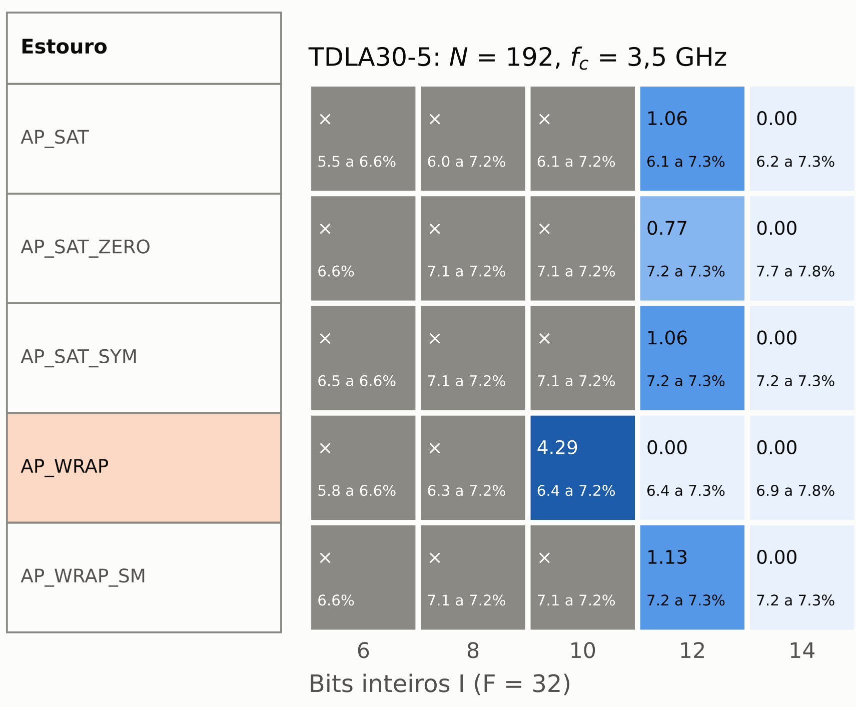
Fundo laranja no rótulo: o modo com o melhor resultado (estouro: pela varredura de I; quantização: pela varredura de F), comparando de menos bits para mais, sem contar a coluna em que tudo é zero

Faixa "x a y": o outro modo muda o valor (a quantização no mapa de I, o estouro no mapa de F); se os dois extremos têm cores diferentes, a célula é dividida na diagonal (em cima, a cor do menor; embaixo, a do maior)

Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo; sem número onde o formato e o modo não foram sintetizados

Perda de SNR (dB)

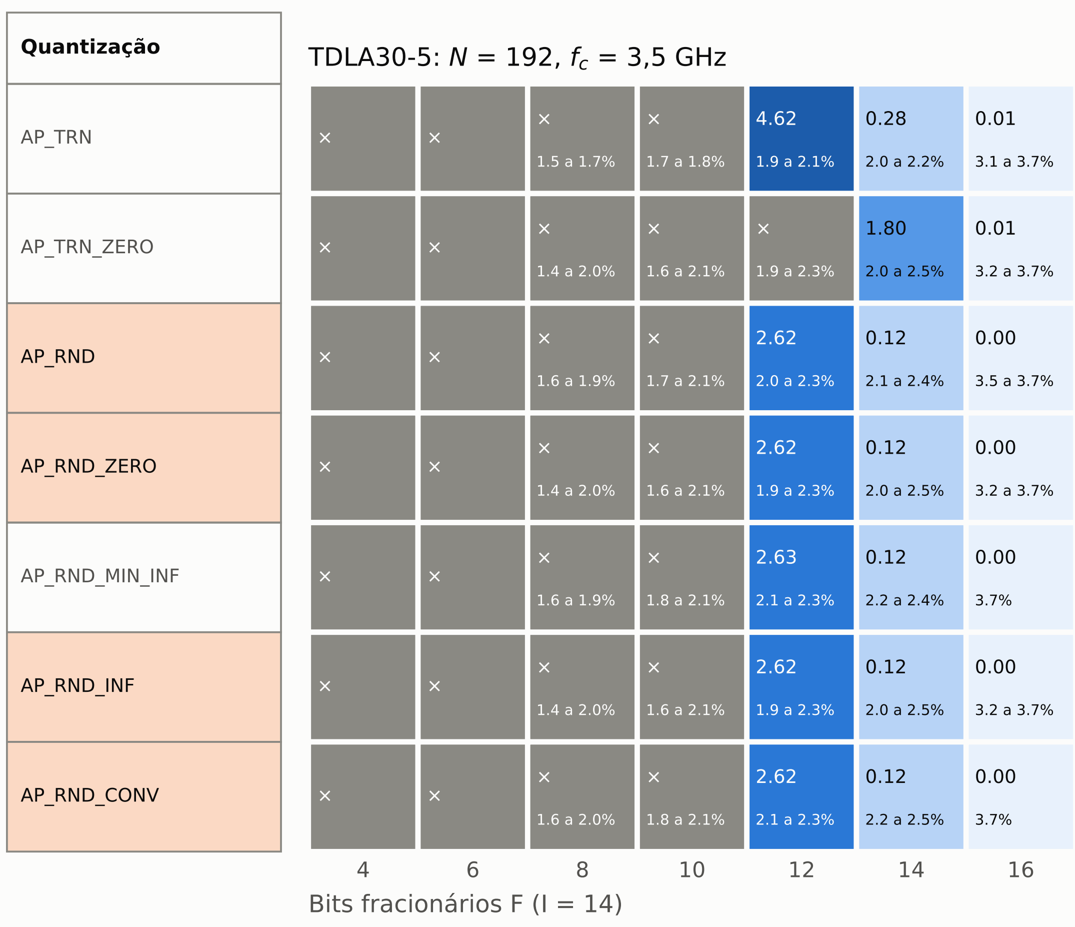
- < 0,1 dB
- 0,1 a 0,5 dB
- 0,5 a 1 dB
- 1 a 2 dB
- 2 a 3 dB
- 3 a 5 dB
- 5 a 10 dB
- > 10 dB
- × não chega a $10^{-2}$
- 🟧 rótulo do melhor modo



TDLA30-5: $N$ = 192, $f_c$ = 3,5 GHz

| Estouro | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|
| AP_SAT | × <br> 5.5 a 6.6% | × <br> 6.0 a 7.2% | × <br> 6.1 a 7.2% | 1.06 <br> 6.1 a 7.3% | 0.00 <br> 6.2 a 7.3% |
| AP_SAT_ZERO | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 0.77 <br> 7.2 a 7.3% | 0.00 <br> 7.7 a 7.8% |
| AP_SAT_SYM | × <br> 6.5 a 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.06 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |
| 🟧 AP_WRAP | × <br> 5.8 a 6.6% | × <br> 6.3 a 7.2% | 4.29 <br> 6.4 a 7.2% | 0.00 <br> 6.4 a 7.3% | 0.00 <br> 6.9 a 7.8% |
| AP_WRAP_SM | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.13 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |

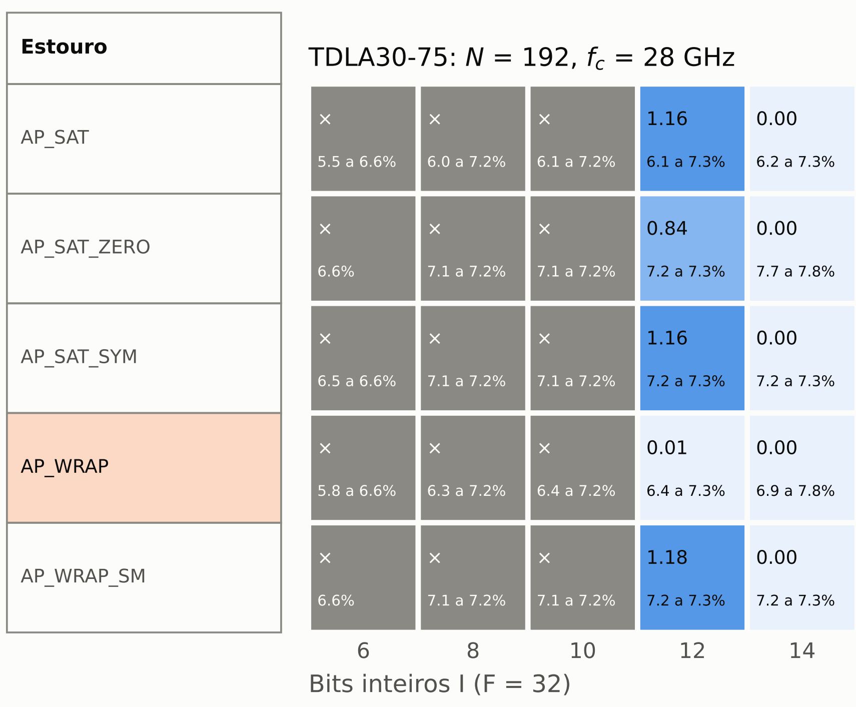
Bits inteiros I (F = 32)



TDLA30-5: $N$ = 192, $f_c$ = 3,5 GHz

| Quantização | 4 | 6 | 8 | 10 | 12 | 14 | 16 |
|---|---|---|---|---|---|---|---|
| AP_TRN | × | × | × <br> 1.5 a 1.7% | × <br> 1.7 a 1.8% | 4.62 <br> 1.9 a 2.1% | 0.28 <br> 2.0 a 2.2% | 0.01 <br> 3.1 a 3.7% |
| AP_TRN_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | × <br> 1.9 a 2.3% | 1.80 <br> 2.0 a 2.5% | 0.01 <br> 3.2 a 3.7% |
| 🟧 AP_RND | × | × | × <br> 1.6 a 1.9% | × <br> 1.7 a 2.1% | 2.62 <br> 2.0 a 2.3% | 0.12 <br> 2.1 a 2.4% | 0.00 <br> 3.5 a 3.7% |
| 🟧 AP_RND_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.62 <br> 1.9 a 2.3% | 0.12 <br> 2.0 a 2.5% | 0.00 <br> 3.2 a 3.7% |
| AP_RND_MIN_INF | × | × | × <br> 1.6 a 1.9% | × <br> 1.8 a 2.1% | 2.63 <br> 2.1 a 2.3% | 0.12 <br> 2.2 a 2.4% | 0.00 <br> 3.7% |
| 🟧 AP_RND_INF | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.62 <br> 1.9 a 2.3% | 0.12 <br> 2.0 a 2.5% | 0.00 <br> 3.2 a 3.7% |
| 🟧 AP_RND_CONV | × | × | × <br> 1.6 a 2.0% | × <br> 1.8 a 2.1% | 2.62 <br> 2.1 a 2.3% | 0.12 <br> 2.2 a 2.5% | 0.00 <br> 3.7% |

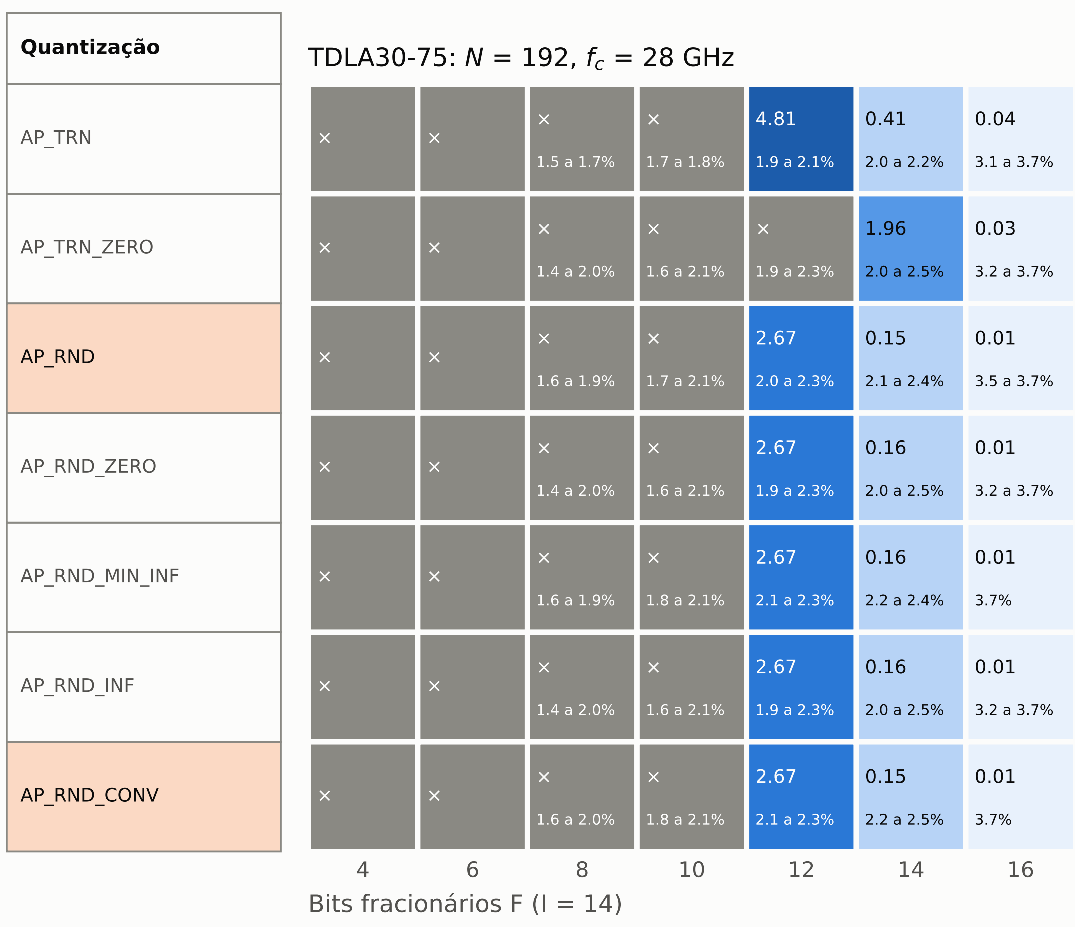
Bits fracionários F (I = 14)



TDLA30-75: $N$ = 192, $f_c$ = 28 GHz

| Estouro | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|
| AP_SAT | × <br> 5.5 a 6.6% | × <br> 6.0 a 7.2% | × <br> 6.1 a 7.2% | 1.16 <br> 6.1 a 7.3% | 0.00 <br> 6.2 a 7.3% |
| AP_SAT_ZERO | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 0.84 <br> 7.2 a 7.3% | 0.00 <br> 7.7 a 7.8% |
| AP_SAT_SYM | × <br> 6.5 a 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.16 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |
| 🟧 AP_WRAP | × <br> 5.8 a 6.6% | × <br> 6.3 a 7.2% | × <br> 6.4 a 7.2% | 0.01 <br> 6.4 a 7.3% | 0.00 <br> 6.9 a 7.8% |
| AP_WRAP_SM | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.18 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |

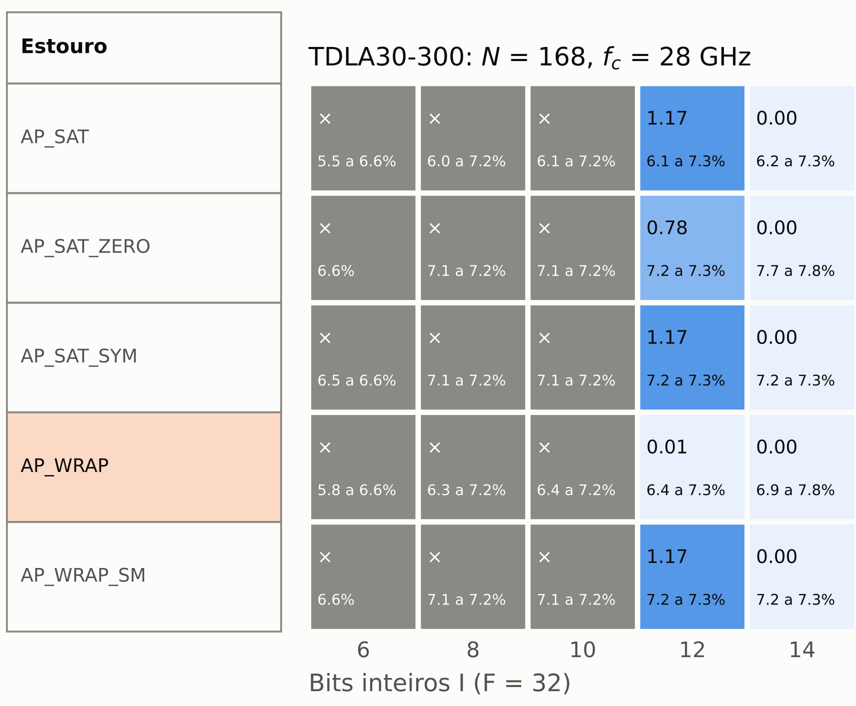
Bits inteiros I (F = 32)



TDLA30-75: $N$ = 192, $f_c$ = 28 GHz

| Quantização | 4 | 6 | 8 | 10 | 12 | 14 | 16 |
|---|---|---|---|---|---|---|---|
| AP_TRN | × | × | × <br> 1.5 a 1.7% | × <br> 1.7 a 1.8% | 4.81 <br> 1.9 a 2.1% | 0.41 <br> 2.0 a 2.2% | 0.04 <br> 3.1 a 3.7% |
| AP_TRN_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | × <br> 1.9 a 2.3% | 1.96 <br> 2.0 a 2.5% | 0.03 <br> 3.2 a 3.7% |
| 🟧 AP_RND | × | × | × <br> 1.6 a 1.9% | × <br> 1.7 a 2.1% | 2.67 <br> 2.0 a 2.3% | 0.15 <br> 2.1 a 2.4% | 0.01 <br> 3.5 a 3.7% |
| AP_RND_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.67 <br> 1.9 a 2.3% | 0.16 <br> 2.0 a 2.5% | 0.01 <br> 3.2 a 3.7% |
| AP_RND_MIN_INF | × | × | × <br> 1.6 a 1.9% | × <br> 1.8 a 2.1% | 2.67 <br> 2.1 a 2.3% | 0.16 <br> 2.2 a 2.4% | 0.01 <br> 3.7% |
| AP_RND_INF | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.67 <br> 1.9 a 2.3% | 0.16 <br> 2.0 a 2.5% | 0.01 <br> 3.2 a 3.7% |
| 🟧 AP_RND_CONV | × | × | × <br> 1.6 a 2.0% | × <br> 1.8 a 2.1% | 2.67 <br> 2.1 a 2.3% | 0.15 <br> 2.2 a 2.5% | 0.01 <br> 3.7% |

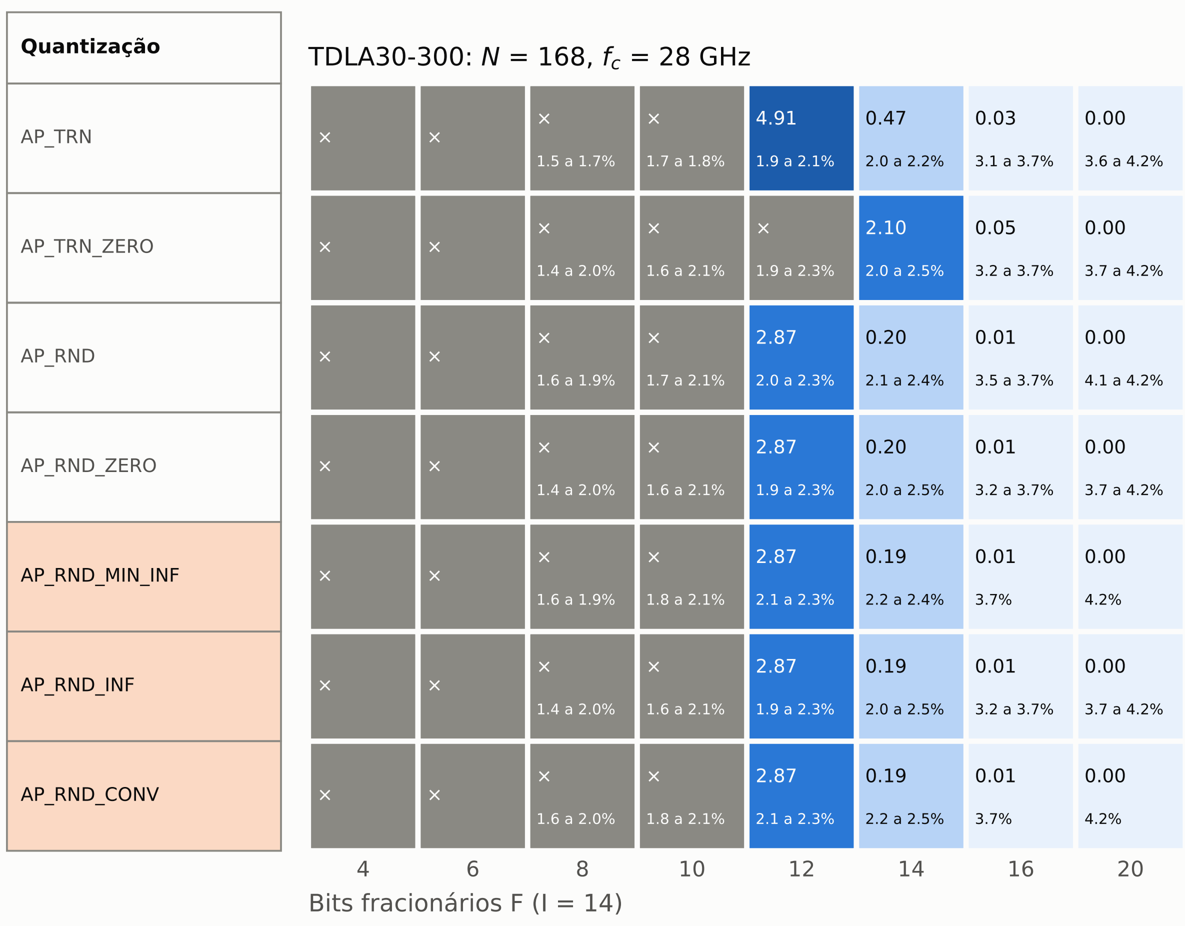
Bits fracionários F (I = 14)



TDLA30-300: $N$ = 168, $f_c$ = 28 GHz

| Estouro | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|
| AP_SAT | × <br> 5.5 a 6.6% | × <br> 6.0 a 7.2% | × <br> 6.1 a 7.2% | 1.17 <br> 6.1 a 7.3% | 0.00 <br> 6.2 a 7.3% |
| AP_SAT_ZERO | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 0.78 <br> 7.2 a 7.3% | 0.00 <br> 7.7 a 7.8% |
| AP_SAT_SYM | × <br> 6.5 a 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.17 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |
| 🟧 AP_WRAP | × <br> 5.8 a 6.6% | × <br> 6.3 a 7.2% | × <br> 6.4 a 7.2% | 0.01 <br> 6.4 a 7.3% | 0.00 <br> 6.9 a 7.8% |
| AP_WRAP_SM | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.17 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |

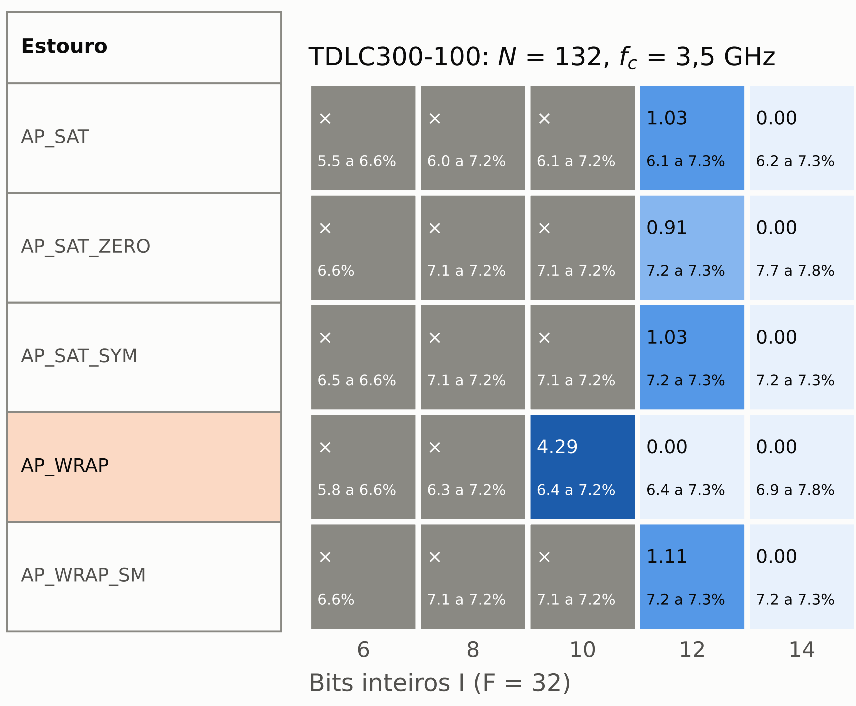
Bits inteiros I (F = 32)



TDLA30-300: $N$ = 168, $f_c$ = 28 GHz

| Quantização | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 20 |
|---|---|---|---|---|---|---|---|---|
| AP_TRN | × | × | × <br> 1.5 a 1.7% | × <br> 1.7 a 1.8% | 4.91 <br> 1.9 a 2.1% | 0.47 <br> 2.0 a 2.2% | 0.03 <br> 3.1 a 3.7% | 0.00 <br> 3.6 a 4.2% |
| AP_TRN_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | × <br> 1.9 a 2.3% | 2.10 <br> 2.0 a 2.5% | 0.05 <br> 3.2 a 3.7% | 0.00 <br> 3.7 a 4.2% |
| AP_RND | × | × | × <br> 1.6 a 1.9% | × <br> 1.7 a 2.1% | 2.87 <br> 2.0 a 2.3% | 0.20 <br> 2.1 a 2.4% | 0.01 <br> 3.5 a 3.7% | 0.00 <br> 4.1 a 4.2% |
| AP_RND_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.87 <br> 1.9 a 2.3% | 0.20 <br> 2.0 a 2.5% | 0.01 <br> 3.2 a 3.7% | 0.00 <br> 3.7 a 4.2% |
| 🟧 AP_RND_MIN_INF | × | × | × <br> 1.6 a 1.9% | × <br> 1.8 a 2.1% | 2.87 <br> 2.1 a 2.3% | 0.19 <br> 2.2 a 2.4% | 0.01 <br> 3.7% | 0.00 <br> 4.2% |
| 🟧 AP_RND_INF | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.87 <br> 1.9 a 2.3% | 0.19 <br> 2.0 a 2.5% | 0.01 <br> 3.2 a 3.7% | 0.00 <br> 3.7 a 4.2% |
| 🟧 AP_RND_CONV | × | × | × <br> 1.6 a 2.0% | × <br> 1.8 a 2.1% | 2.87 <br> 2.1 a 2.3% | 0.19 <br> 2.2 a 2.5% | 0.01 <br> 3.7% | 0.00 <br> 4.2% |

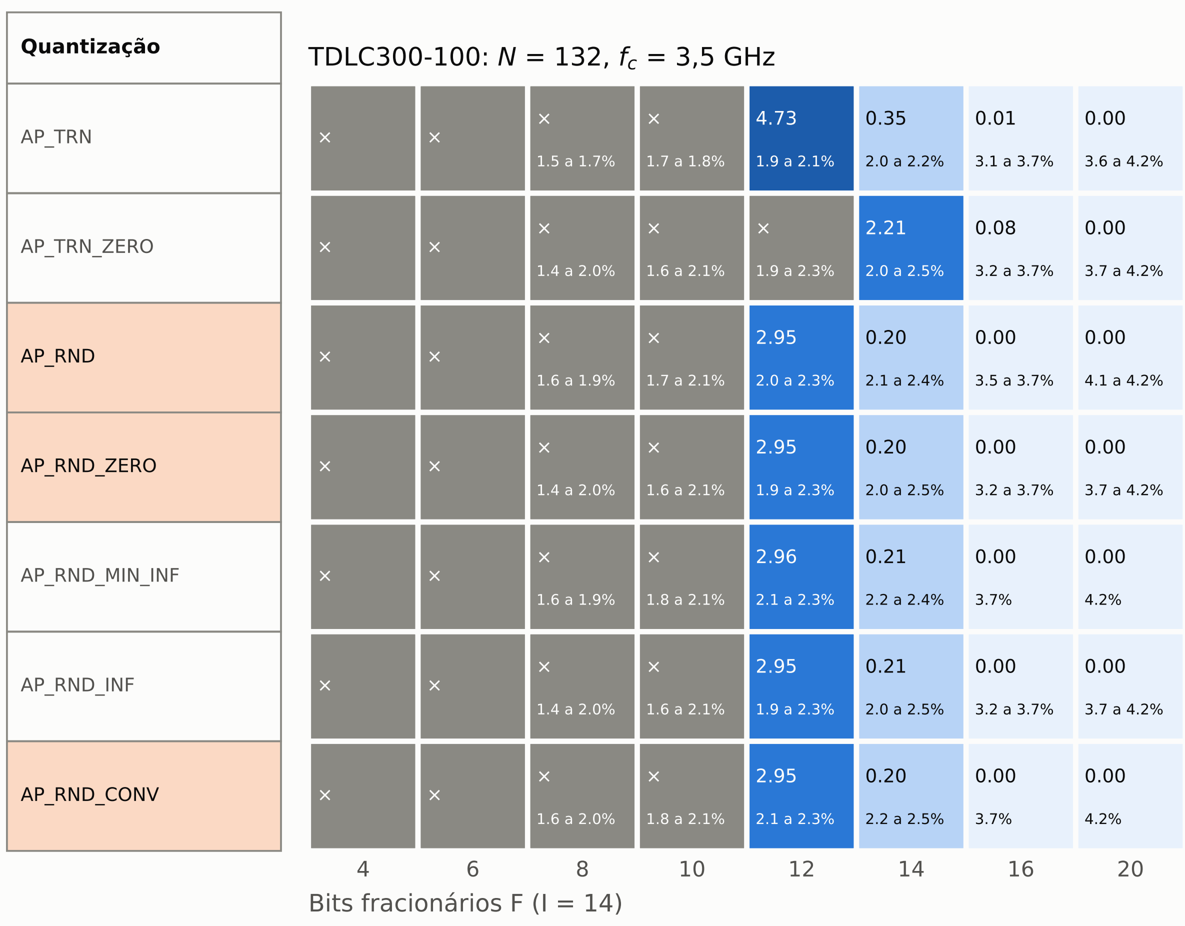
Bits fracionários F (I = 14)



TDLC300-100: $N$ = 132, $f_c$ = 3,5 GHz

| Estouro | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|
| AP_SAT | × <br> 5.5 a 6.6% | × <br> 6.0 a 7.2% | × <br> 6.1 a 7.2% | 1.03 <br> 6.1 a 7.3% | 0.00 <br> 6.2 a 7.3% |
| AP_SAT_ZERO | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 0.91 <br> 7.2 a 7.3% | 0.00 <br> 7.7 a 7.8% |
| AP_SAT_SYM | × <br> 6.5 a 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.03 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |
| 🟧 AP_WRAP | × <br> 5.8 a 6.6% | × <br> 6.3 a 7.2% | 4.29 <br> 6.4 a 7.2% | 0.00 <br> 6.4 a 7.3% | 0.00 <br> 6.9 a 7.8% |
| AP_WRAP_SM | × <br> 6.6% | × <br> 7.1 a 7.2% | × <br> 7.1 a 7.2% | 1.11 <br> 7.2 a 7.3% | 0.00 <br> 7.2 a 7.3% |

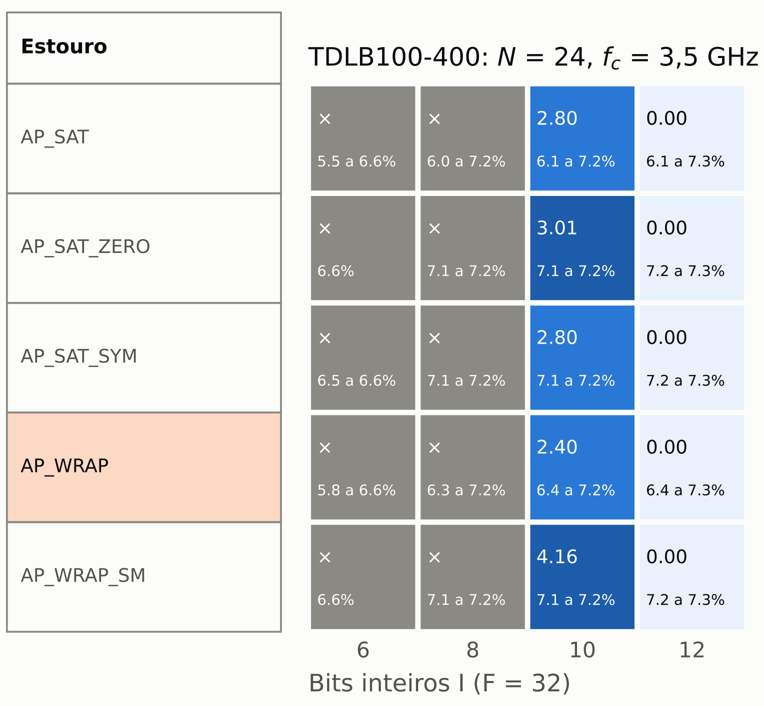
Bits inteiros I (F = 32)



TDLC300-100: $N$ = 132, $f_c$ = 3,5 GHz

| Quantização | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 20 |
|---|---|---|---|---|---|---|---|---|
| AP_TRN | × | × | × <br> 1.5 a 1.7% | × <br> 1.7 a 1.8% | 4.73 <br> 1.9 a 2.1% | 0.35 <br> 2.0 a 2.2% | 0.01 <br> 3.1 a 3.7% | 0.00 <br> 3.6 a 4.2% |
| AP_TRN_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | × <br> 1.9 a 2.3% | 2.21 <br> 2.0 a 2.5% | 0.08 <br> 3.2 a 3.7% | 0.00 <br> 3.7 a 4.2% |
| 🟧 AP_RND | × | × | × <br> 1.6 a 1.9% | × <br> 1.7 a 2.1% | 2.95 <br> 2.0 a 2.3% | 0.20 <br> 2.1 a 2.4% | 0.00 <br> 3.5 a 3.7% | 0.00 <br> 4.1 a 4.2% |
| 🟧 AP_RND_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.95 <br> 1.9 a 2.3% | 0.20 <br> 2.0 a 2.5% | 0.00 <br> 3.2 a 3.7% | 0.00 <br> 3.7 a 4.2% |
| AP_RND_MIN_INF | × | × | × <br> 1.6 a 1.9% | × <br> 1.8 a 2.1% | 2.96 <br> 2.1 a 2.3% | 0.21 <br> 2.2 a 2.4% | 0.00 <br> 3.7% | 0.00 <br> 4.2% |
| AP_RND_INF | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.95 <br> 1.9 a 2.3% | 0.21 <br> 2.0 a 2.5% | 0.00 <br> 3.2 a 3.7% | 0.00 <br> 3.7 a 4.2% |
| 🟧 AP_RND_CONV | × | × | × <br> 1.6 a 2.0% | × <br> 1.8 a 2.1% | 2.95 <br> 2.1 a 2.3% | 0.20 <br> 2.2 a 2.5% | 0.00 <br> 3.7% | 0.00 <br> 4.2% |

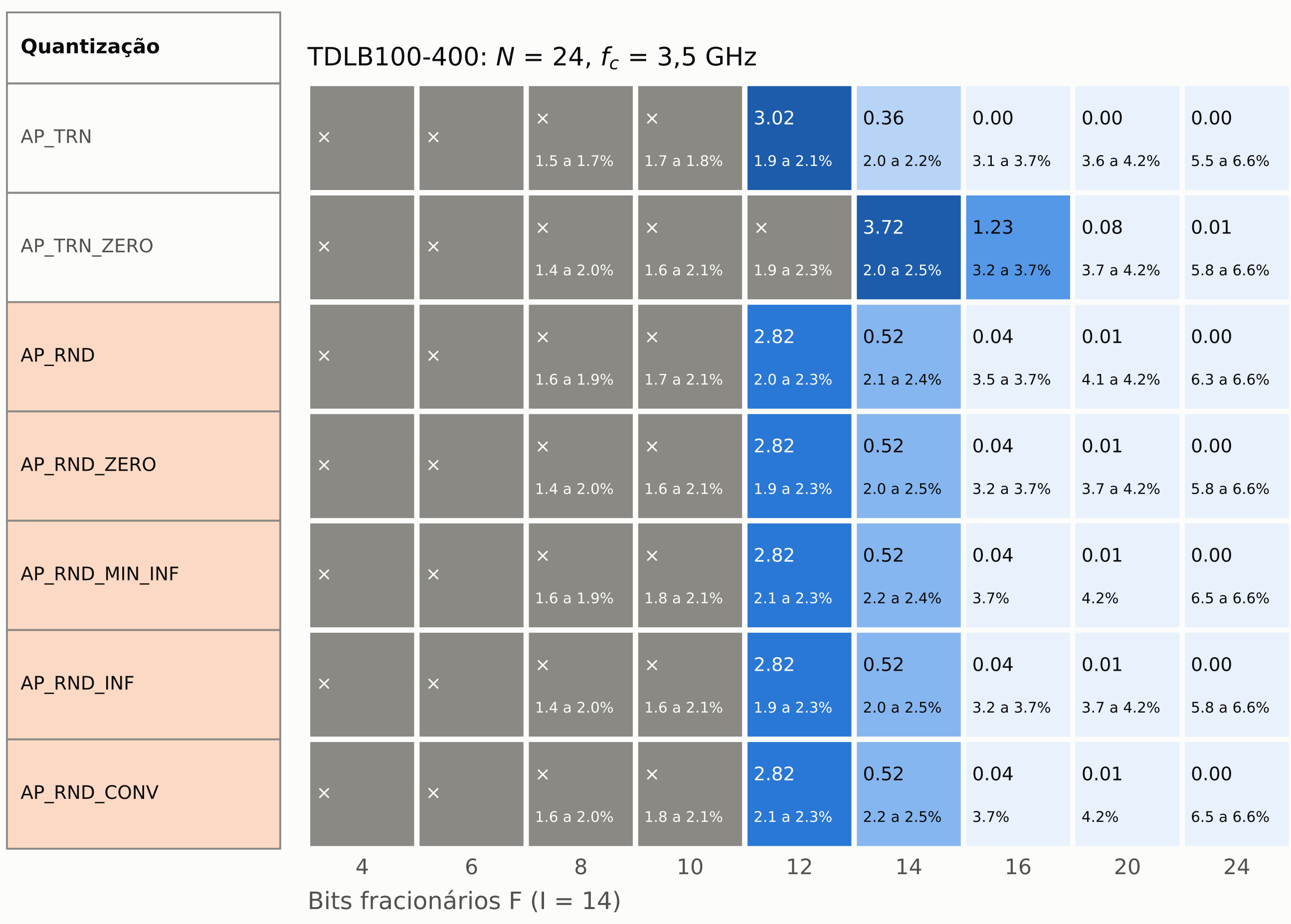
Bits fracionários F (I = 14)



TDLB100-400: $N$ = 24, $f_c$ = 3,5 GHz

| Estouro | 6 | 8 | 10 | 12 |
|---|---|---|---|---|
| AP_SAT | × <br> 5.5 a 6.6% | × <br> 6.0 a 7.2% | 2.80 <br> 6.1 a 7.2% | 0.00 <br> 6.1 a 7.3% |
| AP_SAT_ZERO | × <br> 6.6% | × <br> 7.1 a 7.2% | 3.01 <br> 7.1 a 7.2% | 0.00 <br> 7.2 a 7.3% |
| AP_SAT_SYM | × <br> 6.5 a 6.6% | × <br> 7.1 a 7.2% | 2.80 <br> 7.1 a 7.2% | 0.00 <br> 7.2 a 7.3% |
| 🟧 AP_WRAP | × <br> 5.8 a 6.6% | × <br> 6.3 a 7.2% | 2.40 <br> 6.4 a 7.2% | 0.00 <br> 6.4 a 7.3% |
| AP_WRAP_SM | × <br> 6.6% | × <br> 7.1 a 7.2% | 4.16 <br> 7.1 a 7.2% | 0.00 <br> 7.2 a 7.3% |

Bits inteiros I (F = 32)



TDLB100-400: $N$ = 24, $f_c$ = 3,5 GHz

| Quantização | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 20 | 24 |
|---|---|---|---|---|---|---|---|---|---|
| AP_TRN | × | × | × <br> 1.5 a 1.7% | × <br> 1.7 a 1.8% | 3.02 <br> 1.9 a 2.1% | 0.36 <br> 2.0 a 2.2% | 0.00 <br> 3.1 a 3.7% | 0.00 <br> 3.6 a 4.2% | 0.00 <br> 5.5 a 6.6% |
| AP_TRN_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | × <br> 1.9 a 2.3% | 3.72 <br> 2.0 a 2.5% | 1.23 <br> 3.2 a 3.7% | 0.08 <br> 3.7 a 4.2% | 0.01 <br> 5.8 a 6.6% |
| 🟧 AP_RND | × | × | × <br> 1.6 a 1.9% | × <br> 1.7 a 2.1% | 2.82 <br> 2.0 a 2.3% | 0.52 <br> 2.1 a 2.4% | 0.04 <br> 3.5 a 3.7% | 0.01 <br> 4.1 a 4.2% | 0.00 <br> 6.3 a 6.6% |
| 🟧 AP_RND_ZERO | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.82 <br> 1.9 a 2.3% | 0.52 <br> 2.0 a 2.5% | 0.04 <br> 3.2 a 3.7% | 0.01 <br> 3.7 a 4.2% | 0.00 <br> 5.8 a 6.6% |
| 🟧 AP_RND_MIN_INF | × | × | × <br> 1.6 a 1.9% | × <br> 1.8 a 2.1% | 2.82 <br> 2.1 a 2.3% | 0.52 <br> 2.2 a 2.4% | 0.04 <br> 3.7% | 0.01 <br> 4.2% | 0.00 <br> 6.5 a 6.6% |
| 🟧 AP_RND_INF | × | × | × <br> 1.4 a 2.0% | × <br> 1.6 a 2.1% | 2.82 <br> 1.9 a 2.3% | 0.52 <br> 2.0 a 2.5% | 0.04 <br> 3.2 a 3.7% | 0.01 <br> 3.7 a 4.2% | 0.00 <br> 5.8 a 6.6% |
| 🟧 AP_RND_CONV | × | × | × <br> 1.6 a 2.0% | × <br> 1.8 a 2.1% | 2.82 <br> 2.1 a 2.3% | 0.52 <br> 2.2 a 2.5% | 0.04 <br> 3.7% | 0.01 <br> 4.2% | 0.00 <br> 6.5 a 6.6% |

Bits fracionários F (I = 14)